Group A: By Interacting Object (Box, Sphere, Cylinder) — Motions: Kinematic (Horizontal, Vertical, Circle, Hold) & Dynamic (Drop, Slide, Throw, Tumble)

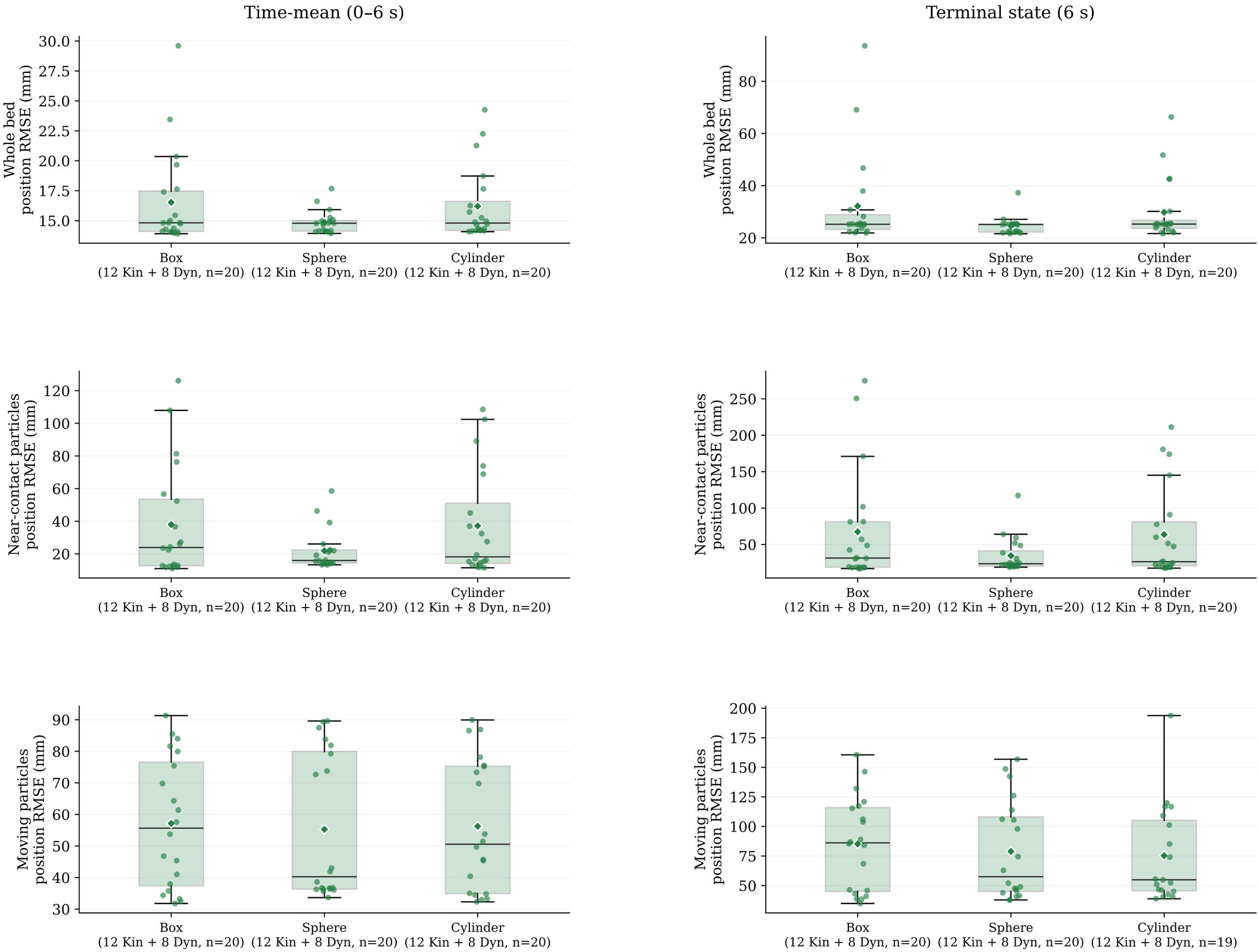

Figure 1a: Frozen object-model checkpoint 11,000. Elementary objects (Box, Sphere, Cylinder), each aggregated across its 20 held-out trajectories: 12 Kinematic (Horizontal, Vertical, Circular, Hold; 3 each) and 8 Dynamic (Drop, Slide, Throw, Tumble; 2 each). Rows show particle-position RMSE for the entire bed, near-contact particles (≤0.05 m), and moving particles (>0.01 m/s).